



TAMPINES MERIDIAN JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION

H2 CHEMISTRY

9729

Paper 1 Multiple Choice

22 September 2023

1 hour

Additional materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Write your name, class and register number on the Answer Sheet in the spaces provided.

There are **thirty** questions in this paper. Answer **all** questions. For each question, there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the use of the Answer Sheet very carefully.

You are advised to fill in the Answer Sheet as you go along; no additional time will be given for the transfer of answers once the examination has ended.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

Use of Answer Sheet

Ensure you have written your name, class register number and class on the Answer Sheet.

Use a **2B** pencil to shade your answers on the Answer Sheet; erase any mistakes cleanly. Multiple shaded answers to a question will not be accepted.

For shading of class register number on the **Answer Sheet**, please follow the given examples:

If your register number is **1**, then shade **01** in the index number column.

If your register number is **21**, then shade **21** in the index number column.

- 1 A sample of boron contains two naturally occurring isotopes, ^{10}B and ^{11}B . This sample has a relative atomic mass value of 10.75.

What is the percentage of the isotope ^{11}B in the sample?

- A 20 B 25 C 75 D 80

Answer: C

Let % abundance of ^{11}B be $m\%$.

$$10.75 = \frac{(m)(11) + (100-m)(10)}{100}$$

$$m = 75$$

- 2 Use of Data Booklet is relevant to this question.

On losing an electron, which particle would have the greatest number of unpaired electrons?

- A C^-
B Fe^{3+}
C Ti^{2+}
D P

Answer: B

C^- forms C : $[\text{He}] 2s^2 2p^2$ (2 unpaired electrons)

Fe^{3+} forms Fe^{4+} : $[\text{Ar}] 3d^4$ (4 unpaired electrons)

Ti^{2+} forms Ti^{3+} : $[\text{Ar}] 3d^1$ (1 unpaired electrons)

P forms P^+ : $[\text{Ne}] 3s^2 3p^2$ (2 unpaired electrons)

- 3 Particle X has a charge of +1 and a proton number, n .

Particle Y has a proton number of $(n+1)$ and is isoelectronic with X.

Which of the following statements correctly describes X and Y?

- A X and Y are isotopes.
B The atoms of X and Y have the same electronic configurations.
C Y has a charge of +1 and same charge density as that of X.
D Y has a charge of +2 and smaller ionic radius than X.



Answer: **D**

	X	Y
Proton number, n	n	n + 1
No of electrons	n – 1	n – 1
Charge	+1	+2

Option A is wrong

X and **Y** have different number of protons. Hence they cannot be isotopes.

Option B is wrong

X and **Y** have different number of protons. **X** atom has n electrons while **Y** has (n + 1) electrons. Hence **X** and **Y** atoms have different full electronic configurations.

Option C is wrong

Y has charge of +2 while it has same number of electrons as that of **X**. Hence, **Y** has smaller ionic radius. This leads to higher charge density.

Option D is correct

Y has charge of +2 while it has same number of electrons as that of **X**. Hence, **Y** has smaller ionic radius.

- 4 The Valence Shell Electron Pair Repulsion (VSEPR) Theory is used to predict the shapes of molecules or polyatomic ions.

Which of the following pairs of species are planar?

- A** F_2O and PF_5
- B** SO_3^{2-} and ClF_3
- C** BrF_4^- and NO_2
- D** ICl_5 and CO_2

Answer: **C**

F_2O	2 bp 1 lp	Bent	planar
PF_5	5 bp 0 lp	Trigonal bipyramidal	Non-planar
SO_3^{2-}	3 bp 1 lp	Trigonal pyramidal	Non-planar
ClF_3	3 bp 2 lp	T-shaped	Planar
BrF_4^-	4 bp 2 lp	Square planar	Planar
NO_2	2 bp 1 lone e^-	Bent	Planar
ICl_5	5 bp 1 lp	Square pyramidal	Non-planar
CO_2	2 bp 0 lp	linear	Planar

[Turn over



5 Which of these phenomena **cannot** be explained by hydrogen bonding?

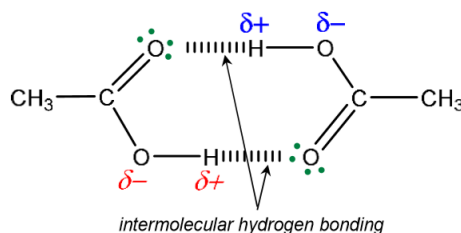
- 1 NH_4Br has a higher boiling point than HBr .
- 2 Ice is less dense than water.
- 3 Dimerisation of NO_2 to form N_2O_4 .
- 4 Ethanoic acid forms dimers which dissolved in organic solvents

A 1, 3 and 4 only B 1 and 3 only C 2 and 4 only D 1 only

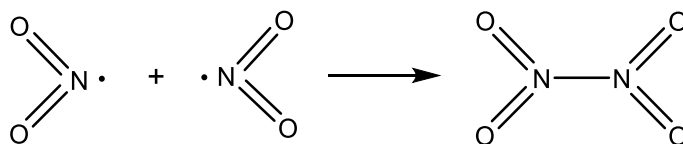
Answer: **B**

Larger amount of energy is needed to overcome the strong ionic bonds in NH_4Br than the weaker intermolecular permanent dipole-permanent dipole attraction in HBr .

When dissolved in organic solvents such as hexane, ethanoic acid form dimers by forming intermolecular hydrogen bonds between two ethanoic acid monomers.



Dimerisation of NO_2 radicals to N_2O_4 .



Ice has a three-dimensional tetrahedral lattice with an open structure held by intermolecular hydrogen bonds. With the open structure, ice has less mass per unit volume (i.e. lower density) than water.

- 6 A 2.50 dm³ vessel containing ethane gas at a pressure of 1.50×10^5 Pa is connected to a 7.50 dm³ vessel containing methane gas at a pressure of 2.50×10^5 Pa. The gases are allowed to mix freely. The temperature of the system is then raised from 100 °C to 350 °C. (Note that there is no chemical reaction between both gases)

What is the final pressure of the system?

- A 2.25×10^5 Pa
B 3.76×10^5 Pa
C 6.68×10^5 Pa
D 7.88×10^5 Pa

Answer: B

$$P_{\text{ethane}} = 1.50 \times 10^5 \times 2.50/10 = 3.75 \times 10^4 \text{ Pa}$$

$$P_{\text{methane}} = 2.5 \times 10^5 \times 7.50/10 = 1.875 \times 10^5 \text{ Pa}$$

$$P_{\text{total}} = 3.75 \times 10^4 + 1.875 \times 10^5 = 2.25 \times 10^5 \text{ Pa @ 100K}$$

$$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$
$$\frac{2.25 \times 10^5}{(100 + 273)} = \frac{p_2}{(350 + 273)}$$

$$p_2 = \underline{\underline{3.76 \times 10^5 \text{ Pa}}}$$

- 7 Which statements concerning only the elements in the third period, sodium to argon, are correct?
- 1 The element with the highest pH for its chloride in water is sodium.
 - 2 The element with the highest electrical conductivity is aluminium.
 - 3 The element with the highest melting point for its oxide is silicon
 - 4 The element that has eight atoms in its molecule is sulfur.
- A 1 and 4 only
B 2 and 3 only
C 1, 2 and 4 only
D 1, 3 and 4 only

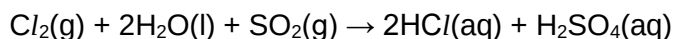
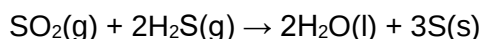
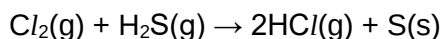
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Answer: **C**

All statements are correct except for option 3. MgO has highest melting point among Period 3 chlorides.

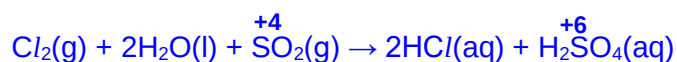
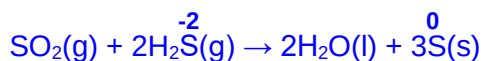
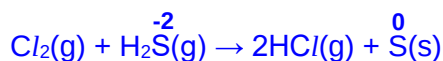
8 The equations for three reactions are given below:



Which is the correct order of strength of the three reacting gases as oxidising agents?

	Strongest	—————→	Weakest
A	Cl_2	SO_2	H_2S
B	Cl_2	H_2S	SO_2
C	H_2S	SO_2	Cl_2
D	SO_2	H_2S	Cl_2

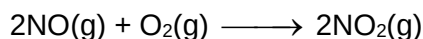
Answer: **A**



From equation (1) and (2), chlorine gas can oxidise both H_2S and SO_2 , thus it is the strongest oxidizing agent.

From equation (1) and (2), hydrogen sulfide get oxidised by SO_2 and HCl , thus it is the weakest oxidising agent.

- 9 The equation for the reaction of nitrogen monoxide with oxygen is shown below.



From initial rate experiments, the following rate equation was derived.

$$\text{Rate} = k[\text{NO}]^2[\text{O}_2]$$

The results of the initial rates experiments are shown.

initial [NO] / mol dm ⁻³	initial [O ₂] / mol dm ⁻³	initial rate of formation of NO ₂ / mol dm ⁻³ s ⁻¹
0.0010	0.002	7.00 × 10 ⁻⁶
0.0020	0.003	a
b	0.004	1.26 × 10 ⁻⁴

What are the missing values **a** and **b**?

	a	b
A	1.9 × 10 ⁻⁵	0.0030
B	1.9 × 10 ⁻⁵	0.0045
C	4.2 × 10 ⁻⁵	0.0030
D	4.2 × 10 ⁻⁵	0.0045

Answer: **C**

$$\text{Rate} = k[\text{NO}]^2[\text{O}_2]$$

Comparing 1st and 2nd experiment

[NO] doubled and [O₂] x1.5

so rate increases 2² = 4 and increases another x1.5.

So total rate increases 4 × 1.5 = 6

$$\Rightarrow \text{Initial rate increases by } 7 \times 10^{-6} \times 6 = 4.2 \times 10^{-5}$$

Comparing 1st and 3rd experiment

Initial rate increased by (1.26 × 10⁻⁴) / (7 × 10⁻⁶) = 18

Since [O₂] doubled, means the rate must have increased by 18/2 = 9 times due to the increase in [NO].

$$\Rightarrow [\text{NO}] \text{ must have increased by } \sqrt{9} = 3 \text{ times}$$

$$\Rightarrow \text{Value of } [\text{NO}] \text{ is } 0.003$$

[Turn over]



- 10 Caffeine is a drug that increases the activity of the brain and nervous system. Its elimination from the body follows a first-order kinetics with a half-life of 5.0 h.

How long will it take for someone who took 50 mg of caffeine, to have 10 mg of caffeine left in his system?

- A 5.0 h B 11.6 h C 25.0 h D 58.0 h

Answer: B

For first order kinetics,
 $t_{1/2} = 5.0$ h

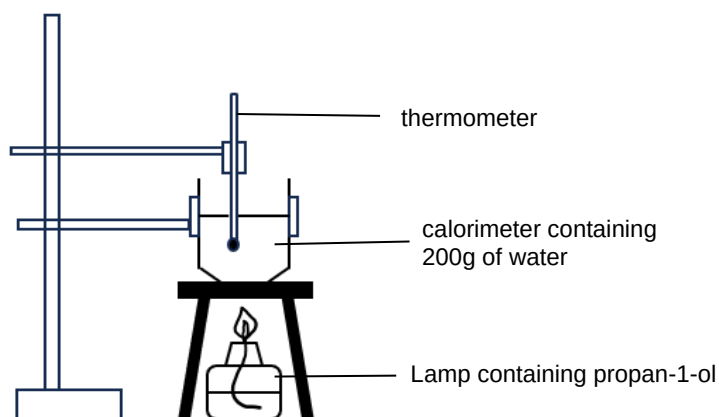
$$[A]_t = [A]_0 \times \left(\frac{1}{2}\right)^n$$

$$10 = 50 \times \left(\frac{1}{2}\right)^n$$

$$\Rightarrow n = 2.32$$

$$\text{time} = 2.32 \times 5 = 11.6 \text{ h}$$

- 11 A student carried out an experiment shown below to determine the enthalpy change of combustion of propan-1-ol ($M_r = 60.0$). It was found that the combustion of 1.0 g of propan-1-ol raises the temperature of 200 g of water by 40 °C.



Given that the enthalpy change of combustion of propan-1-ol is $-2200 \text{ kJ mol}^{-1}$, what is the efficiency of the heat transfer process? Assuming the specific heat capacity of water is $4.18 \text{ J g}^{-1} \text{ K}^{-1}$.

- A 88.5 %
- B 91.2 %
- C 95.2 %
- D 99.5 %

Answer: **B**

$$\text{Heat absorbed by water} = mc\Delta T = (200)(4.18)(40) = 33440\text{J}$$

$$\text{Heat released by combustion of propan-1-ol} = (2200 \times 10^3)(1/60) = 36667\text{J}$$

$$\text{Efficiency} = 33440/36667 = \underline{\underline{91.2\%}}$$

12 For the reaction, $3\text{C(s)} + \text{Cr}_2\text{O}_3\text{(s)} \longrightarrow 2\text{Cr(s)} + 3\text{CO(g)}$ $\Delta H^\circ = +1120 \text{ kJ mol}^{-1}$

Which statement is correct?

- A** The reaction is spontaneous at all temperatures.
- B** The reaction is not spontaneous at any temperature.
- C** The reaction is spontaneous only at low temperature.
- D** The reaction is spontaneous only at high temperature.

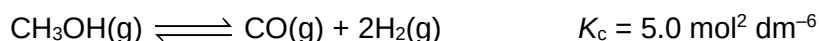
Answer: **D**

$$\Delta G = \Delta H - T\Delta S$$

There is an increase in number of gaseous particles from 0 to 1 mol.

Since $\Delta H > 0$ and $\Delta S > 0$, $\Delta G < 0$ (i.e. spontaneous) only at high temperatures.

13 Gaseous methanol undergoes the following reaction as shown below.



Determine the initial concentration of CH_3OH such that the equilibrium concentration of CH_3OH would be 2.0 mol dm^{-3} .

- A** 3.36
- B** 4.35
- C** 4.68
- D** 5.25

[Turn over



Answer: **A**

Let **a** be the initial concentration of CH₃OH.

	CH ₃ OH(g)	⇌	CO(g)	+	2H ₂ (g)
Initial conc	a		0		0
Change in conc	- x		+ x		+ 2x
Equilibrium conc	2		x		2x

$$K_c = \frac{[H_2]^2[CO]}{[CH_3OH]} = 5$$

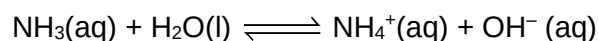
$$K_c = \frac{(2x)^2(x)}{(2)}$$

$$x = 1.357$$

$$a - x = 2$$

$$a = \underline{\underline{3.36}}$$

- 14** Ammonia is commonly used as fertiliser. In an experiment, 0.100 mol dm⁻³ of ammonia dissociates partially in the presence of water according to the equation below.



Which of the following statements is correct?

- A** K_b of ammonia increases upon dilution.
- B** pH remains unchanged upon dilution.
- C** Ammonia acts as a Lewis base.
- D** The degree of dissociation of ammonia remains unchanged upon dilution.

Answer: **C**

Option A: K_b is temperature dependent only.

Option B: Upon dilution, [OH⁻] decreases. pH decreases.

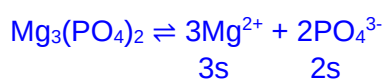
Option C: NH₃ donates an electron pair to form dative bond with H⁺ ion.

Option D: α increases since position of equilibrium shift right upon dilution.



- 15 The numerical value of the solubility product of magnesium phosphate, $\text{Mg}_3(\text{PO}_4)_2$, is 1.0×10^{-24} . What is the solubility of $\text{Mg}_3(\text{PO}_4)_2$?
- A $1.00 \times 10^{-12} \text{ mol dm}^{-3}$
- B $6.21 \times 10^{-6} \text{ mol dm}^{-3}$
- C $1.11 \times 10^{-5} \text{ mol dm}^{-3}$
- D $1.86 \times 10^{-5} \text{ mol dm}^{-3}$

Answer: B



$$K_{\text{sp}} = [\text{Mg}^{2+}]^3 [\text{PO}_4^{3-}]^2$$

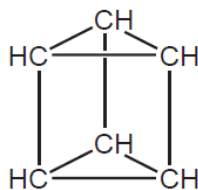
$$1.0 \times 10^{-24} = (3s)^3 (2s)^2$$

$$1.0 \times 10^{-24} = 108s^5$$

$$s = \underline{6.21 \times 10^{-6}}$$

- 16 *Prismane* is an isomer of benzene which has a molecular formula of C_6H_6 , in which one hydrogen atom is attached to each carbon atom. *Prismane* is far less stable than benzene.

How many di-brominated products, excluding stereoisomers, can *Prismane* form via free radical substitution?



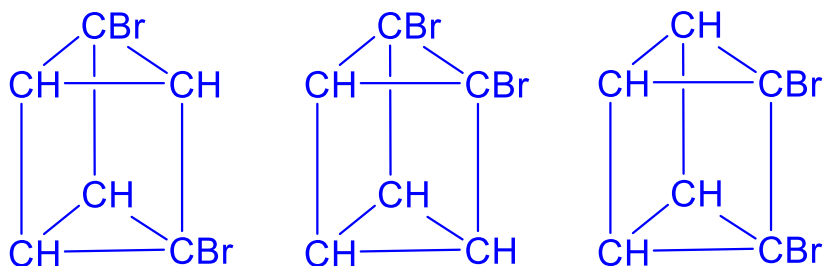
- A 2 B 3 C 4 D 5

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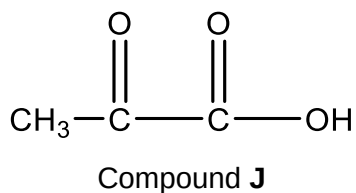


Answer: **B**

3 structural isomers are shown below.



- 17 What is the number of σ and π bonds in compound **K**?



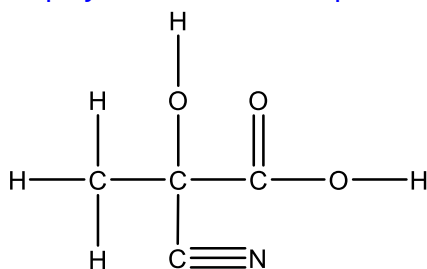
Compound **J** reacts with HCN under suitable conditions to form compound **K**.

What is the number of σ and π bonds in compound **K**?

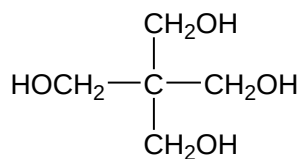
	σ	π
A	9	2
B	12	3
C	13	3
D	14	4

Answer: **B**

The displayed formula of compound **K** is shown below. It has 12 σ and 3 π bonds.



- 18 Pentaerythritol is used as an intermediate in the manufacture of paint.



pentaerythritol

Which statement about pentaerythritol is correct?

- A It can react with HBr (aq) via nucleophilic substitution.
- B Its empirical and molecular formulae are different.
- C It can undergo elimination with hot concentrated sulfuric acid to form an alkene.
- D One mole of pentaerythritol gives two moles of hydrogen gas on reaction with excess sodium.

Answer: D

A is incorrect as pentaerythritol only reacts with HBr gas.

B is incorrect as both the empirical and molecular formulae is $\text{C}_5\text{H}_{12}\text{O}_4$.

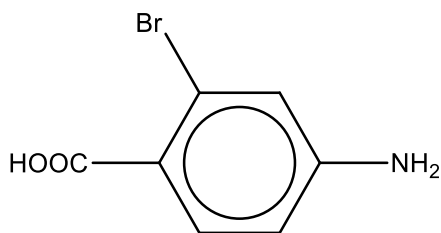
C is incorrect as the central C atom, which is the adjacent C atoms to all the C with $-\text{OH}$ group does not contain any H for elimination to occur.

D is correct. Since 1 mole of $-\text{OH}$ reacts with sodium to form 0.5 mole of hydrogen gas, the presence of 4 $-\text{OH}$ groups in pentaerythritol will produce 2 moles of hydrogen gas.

[Turn over



19 Compound **F** may be synthesised from methylbenzene.

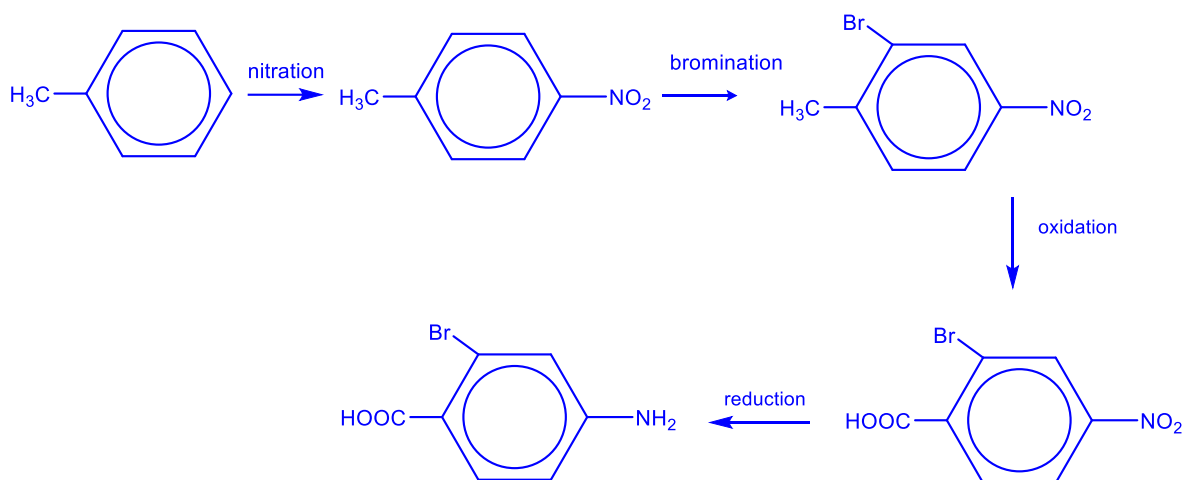


Compound **F**

Which of the following synthetic routes will most likely yield compound **F**?

- A nitration → bromination → oxidation → reduction
- B bromination → oxidation → nitration → reduction
- C bromination → nitration → oxidation → reduction
- D nitration → reduction → bromination → oxidation

Answer: **A**



20 Which of the following could explain why benzene does not undergo electrophilic addition reaction?

- 1 Delocalised pi electrons stabilise the structure.
- 2 Benzene is unable to act as a nucleophile.
- 3 All carbon atoms in benzene are sp^2 hybridised.

A 1,2 and 3 **B** 1 and 2 only **C** 2 and 3 only **D** 1 only

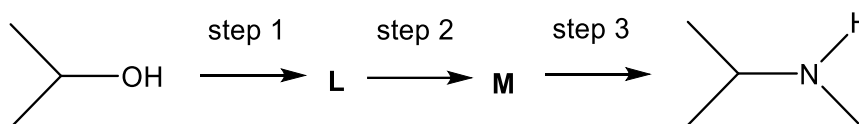
Answer: **D**

1 is correct. The six unbonded p electrons are completely delocalised (free to move throughout) in the π electron cloud above and below the plane of the ring. It is this delocalisation that gives benzene its extra stability, i.e. the benzene ring is *resonance stabilised*.

2 is incorrect. Benzene is electron-rich and can act as a nucleophile.

3 is incorrect. Cannot explain why benzene does not undergo electrophilic addition reaction even though it is a correct statement.

21 The following is a series of reactions involving propan-2-ol.

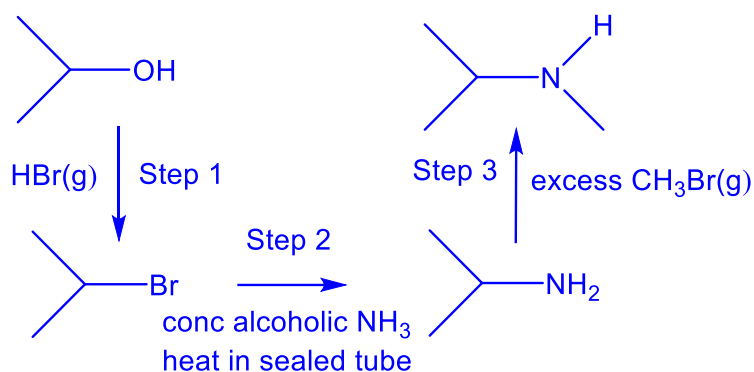


Which of the following give the correct reagents and conditions for steps 1, 2 and 3?

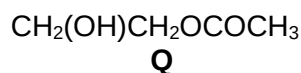
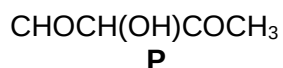
	step 1	step 2	step 3
A	alcoholic KCN, heat	$\text{H}_2(\text{g})$, Pt, high heat and pressure	excess $\text{CH}_3\text{Br}(\text{g})$
B	$\text{HBr}(\text{g})$	conc alcoholic NH_3 , heat in sealed tube	excess $\text{CH}_3\text{Br}(\text{g})$
C	$\text{HBr}(\text{g})$	alcoholic KCN, heat	$\text{H}_2(\text{g})$, Pt, high heat and pressure
D	acidified $\text{K}_2\text{Cr}_2\text{O}_7$, heat	HCN , trace $\text{NaOH}(\text{aq})$, cold	$\text{H}_2(\text{g})$, Pt, high heat and pressure

[Turn over



Answer: **B**

22 Compounds **P** and **Q** have the following formulae:



Which of the following statements apply to these compounds?

- 1 Both react with 2,4–dinitrophenylhydrazine.
- 2 Both react with aqueous alkaline iodine
- 3 **P** reacts with Tollen's reagent but **Q** does not.

- A** 1, 2 and 3 are correct.
- B** 1 and 2 only are correct.
- C** 2 and 3 only are correct.
- D** 3 only is correct.

Answer: **D**

Compound **P** contains aldehyde, methyl ketone and a secondary alcohol.

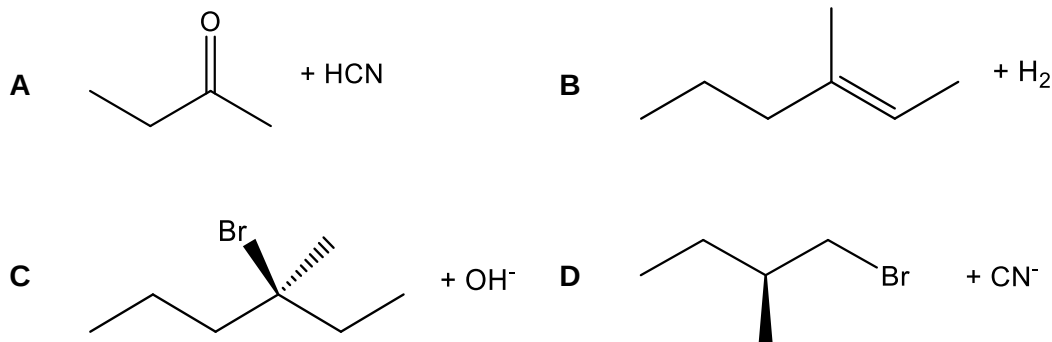
- aldehyde and methyl ketone undergoes condensation with 2,4–DNPH.
- methyl ketone undergoes oxidation with aqueous alkaline iodine.
- aldehyde undergoes oxidation with Tollen's reagent.

Compound **Q** contains primary alcohol and an ester.

There are no aldehyde or ketone, thus no reaction with Tollen's reagent or 2,4–DNPH.

There is no methyl alcohol thus no reaction with aqueous alkaline iodine.

23 Which reaction will **not** produce a mixture of two enantiomers?



Answer: **D**

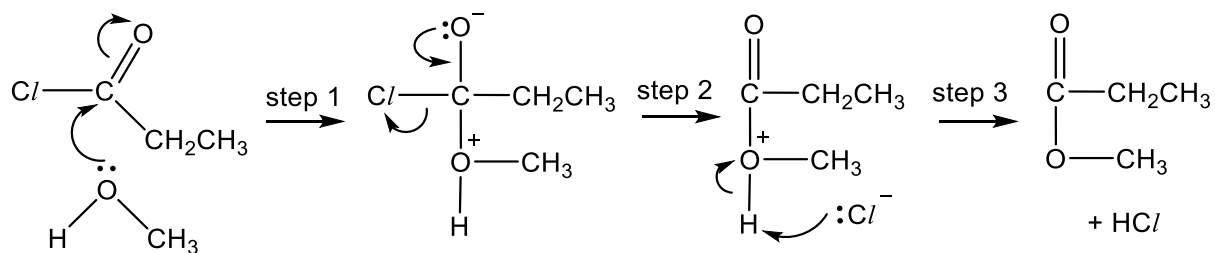
	Reaction	Reasoning	Racemic mixture formed
A	Nucleophilic addition	Trigonal planar sp ² hybridised carbonyl C atom (intermediate)	✓
B	Reduction	Product has 1 chiral carbon	✓
C	Nucleophilic substitution (likely S _N 1)	Trigonal planar sp ² hybridised carbocation (intermediate)	✓
D	Nucleophilic substitution (likely S _N 2)	Pentavalent transition state	X

For reactions A,B and C, there is equal probability of nucleophile attacking trigonal planar carbon atom or carbocation from top or bottom of plane, thus producing equal amounts of enantiomers as products since the carbon atom in product is chiral.

24 The reaction sequence given shows a possible mechanism for the reaction between methanol and propanoyl chloride.

[Turn over





What are the types of reaction occurring in step 1 and 2?

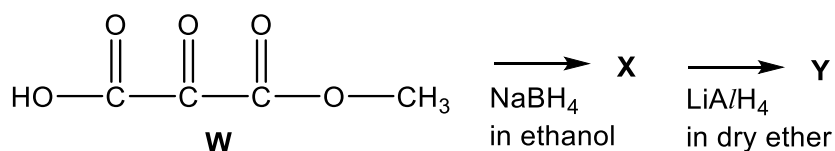
	step 1	step 2
A	electrophilic addition	elimination
B	electrophilic addition	hydrolysis
C	nucleophilic addition	elimination
D	nucleophilic addition	hydrolysis

Answer: **C**

Overall reaction is a nucleophilic acyl substitution which involves a nucleophilic addition to a C=O bond followed by an elimination of a small molecule HCl.

25 The compound **W** shown below undergoes the two reactions as follows:





What could be compounds X and Y?

	X	Y
A	$ \begin{array}{c} \text{O} \quad \text{OH} \quad \text{O} \\ \parallel \quad \quad \parallel \\ \text{HO}-\text{C}-\text{CH}-\text{C}-\text{O}-\text{CH}_3 \end{array} $	$ \begin{array}{c} \text{OH} \quad \text{OH} \quad \text{OH} \\ \quad \quad \\ \text{CH}_2-\text{CH}-\text{CH}_2 \end{array} + \text{HO}-\text{CH}_3 $
B	$ \begin{array}{c} \text{O} \quad \text{OH} \quad \text{O} \\ \parallel \quad \quad \parallel \\ \text{HO}-\text{C}-\text{CH}-\text{C}-\text{O}-\text{CH}_3 \end{array} $	$ \begin{array}{c} \text{OH} \quad \text{OH} \quad \text{OH} \\ \quad \quad \\ \text{CH}_2-\text{CH}-\text{CH}-\text{O}-\text{CH}_3 \end{array} $
C	$ \begin{array}{c} \text{OH} \quad \text{OH} \quad \text{OH} \\ \quad \quad \\ \text{HO}-\text{CH}-\text{CH}-\text{CH}-\text{O}-\text{CH}_3 \end{array} $	$ \begin{array}{c} \text{OH} \quad \text{OH} \quad \text{OH} \\ \quad \quad \\ \text{CH}_2-\text{CH}-\text{CH}_2 \end{array} + \text{HO}-\text{CH}_3 $
D	$ \begin{array}{c} \text{O} \quad \text{OH} \quad \text{OH} \\ \parallel \quad \quad \\ \text{HO}-\text{C}-\text{CH}-\text{CH}_2 \end{array} + \text{HO}-\text{CH}_3 $	$ \begin{array}{c} \text{OH} \quad \text{OH} \quad \text{OH} \\ \quad \quad \\ \text{CH}_2-\text{CH}-\text{CH}_2 \end{array} + \text{HO}-\text{CH}_3 $

Answer: A

Functional groups present: carboxylic acid, ketone, ester.

NaBH₄ reduces only the ketone to form secondary alcohol.

LiAlH₄ subsequently reduces the carboxylic acid into primary alcohol and ester into corresponding alcohols.

- 26 Nitrobenzene reacts with tin in concentrated hydrochloric acid followed by aqueous sodium hydroxide to form phenylamine.

Which of the following statements is correct for the above reaction?

- A Tin is used as a catalyst in this reaction.
- B The oxidation number of the oxygen atom decreased in this reaction.
- C LiAlH₄ is a suitable alternative to carry out this conversion.
- D NaOH is required to form phenylamine from phenylammonium chloride.

Answer: D

[Turn over



Option A incorrect. Sn is oxidised to Sn^{2+} in this reaction; it is not the catalyst but the reducing agent.

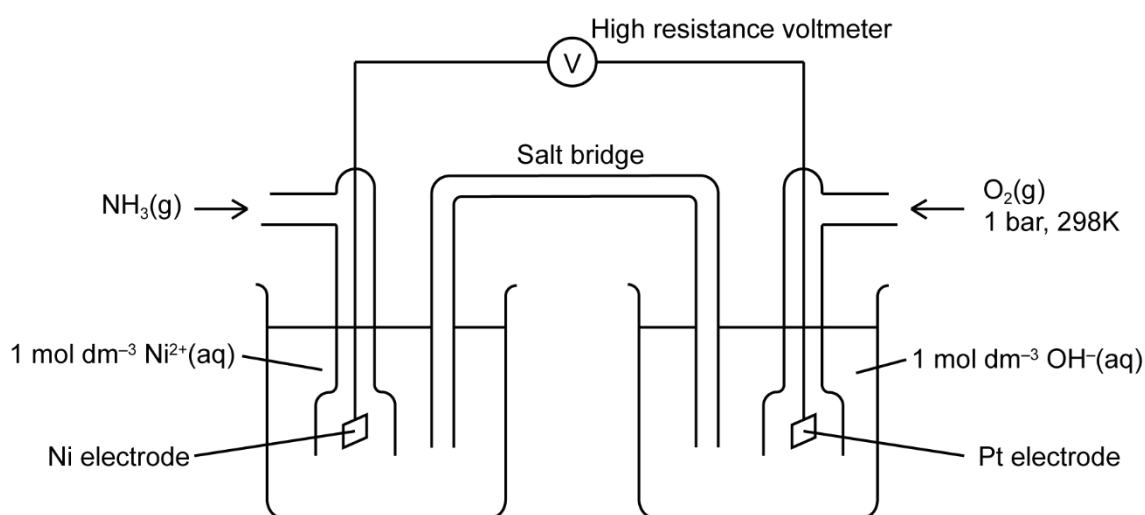
Option B incorrect. The oxidation number of **nitrogen** decreased from +2 in nitrobenzene to -3 in phenylamine.

Option C incorrect. LiAlH_4 cannot be used as it will give rise to other products.

Option D correct. The salt of phenylammonium chloride is formed due to the acid-base reaction between phenylamine formed after reduction of nitrobenzene, with the hydrochloric acid used in the reduction process. The free phenylamine is liberated on the addition of NaOH .

27 Use of Data Booklet is relevant to this question.

An electrochemical experiment was set up, and then ammonia gas passing through the left electrolyte until it reached saturation.

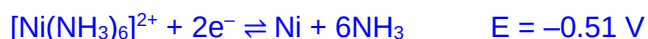


Using the given information and relevant data from the Data Booklet, determine the resultant cell potential.

- A** +0.65 V **B** +0.91 V **C** +1.48 V **D** +1.74 V

Answer: **B**

Relevant equations:



$$E^\circ_{\text{cell}} = (+0.40) - (-0.51) = +0.91 \text{ V}$$

28 An aqueous solution of copper(II) nitrate is electrolysed for 25 min with a constant current of 3.0 A. What is the expected gain in mass of the cathode?



- A 0.025 g B 0.68 g C 1.5 g D 3.0 g

Answer: C

$$\text{Mass} = (25 \times 60) \times 3.0 \div (96500 \times 2) \times 63.5 = 1.5 \text{ g}$$

29 Which statement correctly describes the difference between magnesium metal and titanium metal?

- 1 For two identical samples, the mass of magnesium is less than the mass of titanium.
- 2 Magnesium melts at a lower temperature than titanium.
- 3 Titanium has catalytic properties while magnesium does not.

- A 1, 2 and 3 B 1 and 2 only C 1 only D 3 only

Answer: A

This question is meant to contrast between s-block metals (exemplified by magnesium) and transition metals (exemplified by titanium)

- 1: Transition metals are denser than s-block metals
- 2: Transition metals have higher melting points than s-block metals
- 3: Transition metals has catalytic properties

30 Which of the following processes does **not** involve a change in colour of the solution?

- A Adding Zn(s) to $\text{V}(\text{NO}_3)_2(\text{aq})$
- B Adding $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$ to $\text{I}_2(\text{aq})$
- C Adding concentrated HCl to $\text{Cu}(\text{NO}_3)_2(\text{aq})$
- D Adding methanoic acid to acidified KMnO_4 .

Answer: A

- A: no redox
B: redox
C: ligand exchange
D: oxidation

[Turn over



1	2	3	4	5	6	7	8	9	10
C	B	D	C	B	B	C	A	C	B
11	12	13	14	15	16	17	18	19	20
B	D	A	C	B	B	B	D	A	D
21	22	23	24	25	26	27	28	29	30
B	D	D	C	A	D	B	C	A	A

